QUBO Formulation: Weighted Set Packing Problem

Weighted Set Packing Problem

Problem Statement. Let U be a finite universe of elements and let $\mathcal{S} = \{S_0, S_1, \ldots, S_{m-1}\}$ be a family of subsets of U, where each subset $S_j \subseteq U$ is associated with a non-negative weight $w_j \geq 0$. The Weighted Set Packing Problem seeks to select a subfamily of pairwise disjoint subsets $\mathcal{S}' \subseteq \mathcal{S}$ such that the total weight of the selected subsets is maximized. Formally, the goal is to find a collection of indices $J \subseteq \{0, 1, \ldots, m-1\}$ satisfying $S_j \cap S_k = \emptyset$ for all distinct $j, k \in J$, that maximizes $\sum_{j \in J} w_j$.

Decision Variables. For each subset index $j \in \{0, 1, \ldots, m-1\}$, introduce a binary decision variable $x_j \in \{0, 1\}$. The variable x_j takes the value 1 if subset S_j is selected, and 0 otherwise. The vector $x = (x_0, x_1, \ldots, x_{m-1}) \in \{0, 1\}^m$ encodes a candidate solution.

QUBO Derivation. Step 1 — Objective. The original problem is a maximization:

$$\max_{x \in \{0,1\}^m} \sum_{j=0}^{m-1} w_j x_j. \tag{1}$$

To cast this as a QUBO minimization, we negate the objective:

$$\min_{x \in \{0,1\}^m} \left(-\sum_{j=0}^{m-1} w_j x_j \right). \tag{2}$$

Step 2 — Constraints. The pairwise disjointness requirement is equivalent to enforcing that for every element $e \in U$, at most one selected subset contains e. This yields the following family of linear constraints:

$$\sum_{j : e \in S_j} x_j \leq 1, \quad \forall e \in U. \tag{3}$$

Step 3 — Penalty Terms. Since the variables x_j are binary, the constraint $\sum_{j:e \in S_j} x_j \leq 1$ is violated if and only if there exist distinct indices j, k such that $e \in S_j$, $e \in S_k$, and $x_j = x_k = 1$. Thus, a violation for element e is detected by the product $x_j x_k$ for any pair

$j < k$ sharing element e. We introduce a penalty weight $\lambda > 0$ and construct the penalty function:

$$P(x) = \lambda \sum_{e \in U} \sum_{j<k : e \in S_j \cap S_k} x_j x_k. \tag{4}$$

Expanding the sum over elements, the coefficient of a specific product $x_j x_k$ (with $j < k$) is λ multiplied by the number of elements shared by S_j and S_k. Let $|S_j \cap S_k|$ denote the cardinality of the intersection. The penalty can be rewritten as a sum over all pairs:

$$P(x) = \lambda \sum_{0 \le j < k \le m-1} |S_j \cap S_k| \, x_j x_k. \tag{5}$$

Step 4 — QUBO Hamiltonian. Combining the negated objective and the penalty term, the QUBO Hamiltonian $H(x)$ is:

$$H(x) = -\sum_{j=0}^{m-1} w_j x_j + \lambda \sum_{0 \le j < k \le m-1} |S_j \cap S_k| \, x_j x_k. \tag{6}$$

Step 5 — Q Matrix and Offset. The Hamiltonian is expressed in the standard QUBO form $H(x) = \sum_{j \le k} Q_{j,k} x_j x_k + c$. Reading off the coefficients from the general expression above, we obtain:

$$Q_{j,j} = -w_j, \quad \forall j \in \{0, \ldots, m-1\}, \tag{7}$$

$$Q_{j,k} = \lambda \, |S_j \cap S_k|, \quad \forall 0 \le j < k \le m-1, \tag{8}$$

$$c = 0. \tag{9}$$

Penalty Weight Justification. The penalty weight λ must be chosen sufficiently large to ensure that any constraint violation incurs a cost exceeding the maximum possible gain from the objective. Specifically, let $W_{\max} = \max_j w_j$. If a solution x violates a constraint, there exists at least one pair j, k with $x_j = x_k = 1$ and $S_j \cap S_k \neq \emptyset$. The penalty contribution is at least λ. By removing one of the conflicting variables from an infeasible solution, we obtain a feasible solution whose weight is reduced by at most $W_{\max}$. Therefore, choosing

$$\lambda > W_{\max} \tag{10}$$

ensures that the energy of any infeasible solution is strictly greater than that of a corresponding feasible solution, guaranteeing that the global minimum of $H(x)$ corresponds to a feasible solution.

Correctness Argument. Minimizing $H(x)$ over $\{0,1\}^m$ recovers the optimal set packing solution. For any feasible solution x, the penalty term vanishes, so $H(x) = -\sum_j w_j x_j$, which is minimized exactly when the total weight is maximized. For any infeasible solution x, the penalty $P(x) \ge \lambda$. Since $\lambda > W_{\max}$, the energy increase due to the penalty outweighs any potential reduction in the objective value relative to a feasible solution. Consequently, the global minimum of $H(x)$ must occur at a feasible point, and among feasible points, the minimum energy corresponds to the maximum weight packing.

Illustrative Example. Consider a small instance with $m = 3$ subsets and universe $U = \{a, b\}$. Define the subsets and weights as:

$$S_0 = \{a\}, \quad w_0 = 5; \qquad S_1 = \{a, b\}, \quad w_1 = 8; \qquad S_2 = \{b\}, \quad w_2 = 4. \tag{11}$$

The intersections are $S_0 \cap S_1 = \{a\}$, $S_1 \cap S_2 = \{b\}$, and $S_0 \cap S_2 = \emptyset$. Thus $|S_0 \cap S_1| = 1$, $|S_1 \cap S_2| = 1$, and $|S_0 \cap S_2| = 0$. Choosing $\lambda = 9$ (since $\max w_j = 8$), the Hamiltonian becomes:

$$H(x) = -5x_0 - 8x_1 - 4x_2 + 9x_0x_1 + 9x_1x_2. \tag{12}$$

The Q matrix entries are:

$$Q_{0,0} = -5, \qquad Q_{0,1} = 9, \qquad Q_{0,2} = 0, \tag{13}$$

$$Q_{1,1} = -8, \qquad Q_{1,2} = 9, \qquad Q_{2,2} = -4. \tag{14}$$

The feasible solutions are $x \in \{(0,0,0), (1,0,0), (0,1,0), (0,0,1), (1,0,1)\}$. Evaluating $H(x)$:

$$H(0,0,0) = 0, \qquad H(1,0,0) = -5, \qquad H(0,1,0) = -8, \tag{15}$$

$$H(0,0,1) = -4, \qquad H(1,0,1) = -9. \tag{16}$$

The minimum energy is -9 at $x^* = (1,0,1)$, corresponding to selecting subsets S_0 and S_2 with total weight $5 + 4 = 9$. This is optimal since selecting S_1 alone yields weight 8, which is less than 9.